

C6-3.

$$1. \Delta I(t) = I(t) - \langle I(t) \rangle$$

$$I(t) = \Delta I(t) + \langle I(t) \rangle$$

$$I(t+\tau) = \Delta I(t+\tau) + \langle I(t+\tau) \rangle$$

$$g^{(2)}(\tau) = \frac{(\langle \Delta I(t) \Delta I(t+\tau) \rangle)(\langle I(t+\tau) \rangle + \Delta I(t+\tau))}{\langle I(t) \rangle \langle I(t+\tau) \rangle}$$

$$= \frac{\langle \Delta I(t) \Delta I(t+\tau) \rangle + \langle \Delta I(t) \langle I(t+\tau) \rangle \rangle + \langle \langle I(t) \rangle \Delta I(t+\tau) \rangle + \langle \langle I(t) \rangle \langle I(t+\tau) \rangle \rangle}{\langle I(t) \rangle \langle I(t+\tau) \rangle}$$

$$= \frac{\langle \Delta I(t) \Delta I(t+\tau) \rangle + \langle \Delta I(t) \rangle \langle I(t+\tau) \rangle + \langle \Delta I(t+\tau) \rangle \langle I(t) \rangle + \langle I(t) \rangle \langle I(t+\tau) \rangle}{\langle I(t) \rangle \langle I(t+\tau) \rangle}$$

$$= 1 + \frac{\langle \Delta I(t) \Delta I(t+\tau) \rangle}{\langle I(t) \rangle \langle I(t+\tau) \rangle}$$

$$\text{则 } g^{(2)}(0) = 1 + \frac{\langle (\Delta I(t))^2 \rangle}{\langle I(t) \rangle^2} > 1, \text{ 得证.}$$

$$2. \begin{cases} \hat{b}_1 = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2) \\ \hat{b}_2 = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2) \end{cases}$$

$$\text{则 } \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}$$

$$\begin{cases} \hat{b}'_1 = \hat{b}_1 \cdot e^{i\phi} \\ \hat{b}'_2 = \hat{b}_2 \end{cases} \quad \text{则 } \begin{pmatrix} \hat{b}'_1 \\ \hat{b}'_2 \end{pmatrix} = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix}$$

$$\begin{cases} \hat{c}_1 = \frac{1}{\sqrt{2}}(\hat{b}_1' + \hat{b}_2') \\ \hat{c}_2 = \frac{1}{\sqrt{2}}(\hat{b}_1' - \hat{b}_2') \end{cases} \quad \text{则} \quad \begin{pmatrix} \hat{c}_1 \\ \hat{c}_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{b}_1' \\ \hat{b}_2' \end{pmatrix}$$

$$\begin{pmatrix} \hat{c}_1 \\ \hat{c}_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{i\phi} & 1 \\ e^{i\phi} & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{i\phi} + 1 & e^{i\phi} - 1 \\ e^{i\phi} - 1 & e^{i\phi} + 1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}$$

$$n_3 = \hat{c}_1^\dagger \hat{c}_1, \quad \text{则} \quad n_3 = \frac{1}{2} [\hat{a}_1 (e^{-i\phi} + 1) + \hat{a}_2 (e^{-i\phi} - 1)] \cdot \frac{1}{2}$$

$$[\hat{a}_1 (e^{i\phi} + 1) + \hat{a}_2 (e^{i\phi} - 1)]$$

$$= \frac{1}{2} n_1 (1 + \cos\phi) + \frac{1}{2} n_2 (1 - \cos\phi)$$

$$n_4 = \hat{c}_2^\dagger \hat{c}_2, \quad \text{则} \quad n_4 = \frac{1}{2} n_1 (1 - \cos\phi) + \frac{1}{2} n_2 (1 + \cos\phi)$$

选择题:

1. d 2. a 3. d 4. a 5. d 6. acd

7. a 8. a 9. abd 10. cd